Digna Sukeerthi Abisha David

Embedded Systems Engineer
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Summary

Experienced Embedded Developer with **5+ years of professional experience** in low-level firmware development, hardware software integration, and diagnostics for complex embedded systems. Highly proficient in **C and Embedded C**, with foundational knowledge of C++. Expert in developing low-level controllers, handling interrupt routines, and working directly with peripheral communication protocols (UART, I2C, SPI). Seeking to contribute deep expertise in C/C++ development and performance optimization to a team focused on revolutionizing smart home/IoT devices.

Education

Master of Science (MSc) in Automatic Control, Cybernetics, and Robotics (Specialization: Intelligent Decision Systems and Robotics)
Gdańsk University of Technology, Gdańsk, Poland (Currently Pursuing)

Bachelor of Engineering (BE) in Electronics and Communication
Anna University Chennai, India | May 2018 | Cumulative Grade Point Average (CGPA): 7.11

Professional Experience

Software Engineer | Capgemini, Formerly Aricent/Altran, Chennai, India | *January 2020 – September 2024*

- Key developer during the **SDK Migration** in AT&T project for high-volume Residential Gateways (BGW210, BGW320, CGW450) based on **Broadcom chipsets**. Focused on designing and implementing core low-level functionality and solving critical platform bugs.

Key Contributions & Senior Level Tasks:

- **C/Embedded C Low-Level Development:** Authored, analyzed, and updated robust low-level C code for embedded systems, focusing on bootloaders (U-Boot/CFE), device tree configuration, and memory analysis on **aarch64** architectures.
- **Low-Level Controllers & Interrupts:** Developed and verified controller functionality for peripherals including **GPIO, SFP's, LED's, Watchdog, and EMMC**. Possesses a better understanding of **interrupt handling** and experience in **Character Device Driver** development.
- **Diagnostics & Hardware Troubleshooting:** Proficiently utilized **JTAG Debugger** to debug and resolve critical, platform-related bugs. Confirmed **BIST Functionality** across all products and has the **ability to read schematics and datasheets**.
- **Secure System Implementation:** Implemented a complex security layer involving **AES encryption during compilation** and integrated SecureOS to enable access for **OPTEE, RPMB, and ROOTFS Encryption**.
- **Protocols & Toolchain:** Solid experience with **UART, I2C, and SPI communication protocols**. Proficient in **Buildroot configuration**, Cross-Compilation, and toolchain management.

Trainee (Internship) | Capgemini, Formerly Aricent/Altran, Chennai, India | *February 2019 – January 2020*

- Gained practical experience in software development lifecycle, code review processes, and core embedded system concepts by shadowing senior team members and assisting with entry-level tasks.

Core Technologies & Tools

- **Programming Languages:** C, Embedded C, C++ (Foundational), Python, Shell Scripting
- **Low-Level/Peripherals:** UART, I2C, SPI, GPIO, EMMC, SFP's, Watchdog, Interrupt Handling, Character Device Drivers, PCIe Drivers, Low-Level Controllers
- **SoC Architecture:** Experience with Broadcom Chipsets (BCM6858, BCM6813, BCM63148), **aarch64** architecture, Reading Schematics/Datasheets
- **RTOS/Real-Time (Implied): B**ootloaders (U-Boot/CFE), **Secure Storage (OPTEE, RPMB)**, Linux OS concepts, Buildroot configuration
- **Tools & Methodology:** JTAG Debugger, Vim, Putty, MobaXterm, GIT, Jira, SDLC, AGILE Methodologies, Unit Testing

Awards & Recognition

- **Individual Excellence Award:** Recognized with an individual excellence award for demonstrating accountability in performance reviews.
- **Analytical Skill Commendation:** Commended for rapid assimilation of new modules and strong analytical capabilities in addressing field-related issues.

Languages

- English (Advanced) | Tamil (Advanced) | Polish (Beginner)